

MAMBO-VLA

Multi-modal Action Manipulation for Bimanual Operations via VLA

Two SO-101 arms / One learned policy

One instruction-switched visuomotor policy drives a dinner-table relay — pick, table set-down with handover, place — from language plus four camera views. Trained locally, exported to FP32, benchmarked on Intel CPU.

3/10

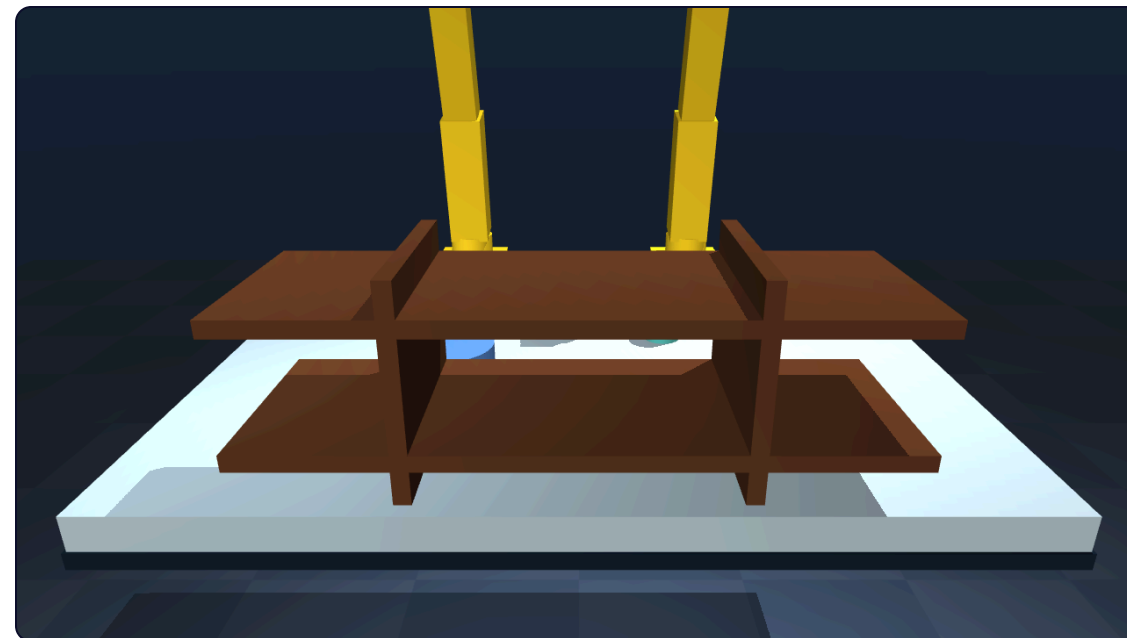
FROZEN SEEDS

40.80 ms

MEDIAN LATENCY

60

DEMOS

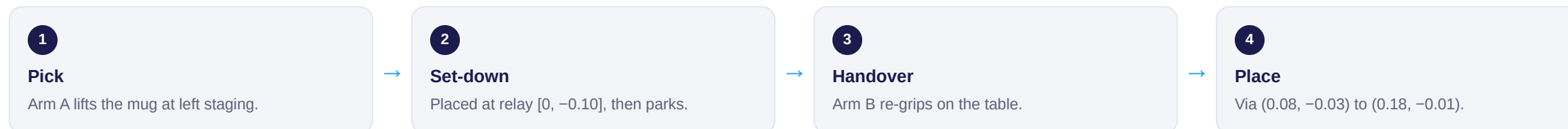


Wide shot · dual-arm table scene · frozen layout shared by training, evaluation, and video

Delivery deck · September 2026 · Scored vehicle **act_full_v3_100k** · 60 demonstrations · All failures retained

Table-supported relay, never airborne

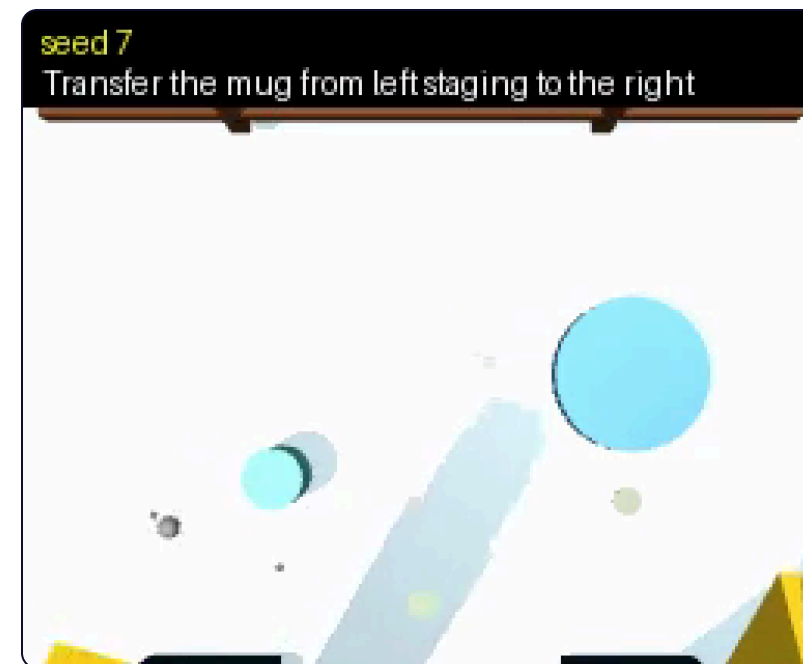
Arm A picks, sets down at the relay point, and parks. Arm B re-grips and places — in two arc legs with a midpoint set-down.



“Transfer the mug from left staging to the right zone via the table relay.”

Bounded grammar: this sentence runs. Anything else is refused with **0 steps** — the refusal is the swap-instruction control (0/10).

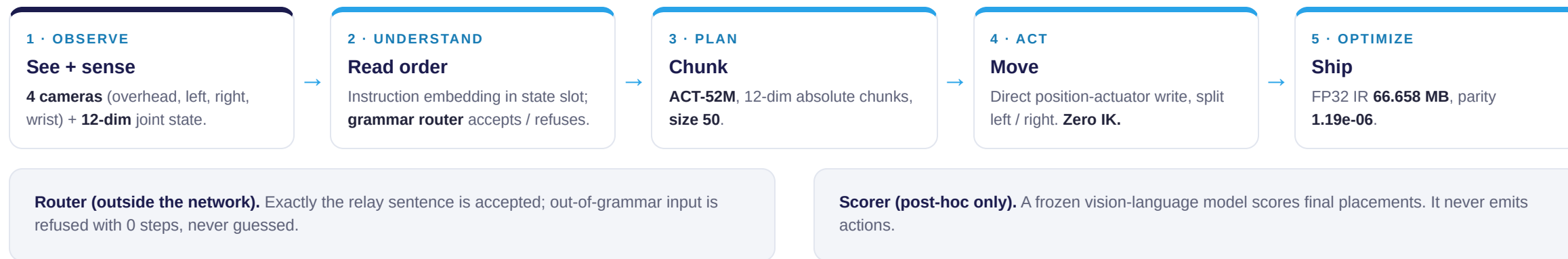
- **Shelf-only scene, no drawer.** 12-dim absolute joint-plus-jaw chunks.
- **4 × 256 × 256 RGB @ 20 fps** plus measured joint state — identical in training, evaluation, and video.



Start state · seed-frozen staging · 768 px capture

One checkpoint, five stages

LangACT is the sole action-generating network — one checkpoint for all commands, switched by sentence, not by weights.



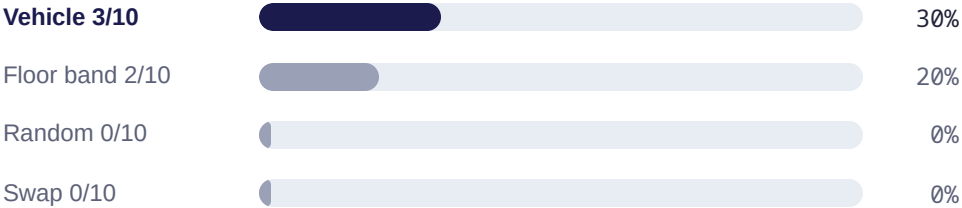
3 / 10 — with controls and floor retained

3/10

Vehicle act_full_v3_100k, 60 demos. Inputs hashed before running; no seed re-rolled; logs and clips kept.

Successes at 7,650 · 7,775 · 7,750 steps (seeds 1, 4, 7). Best failing seed moved 2,457 mm.

HONEST BARS • SAME DENOMINATOR (10)



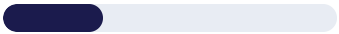
Floor: prior series 2/10 · 0/10 · 1/10, reported not hidden. Random: 0 mm all seeds. Swap: router refusal, 0 steps.

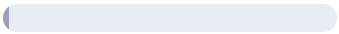
SEED	RESULT	STEPS
1	SUCCESS	7650
4	SUCCESS	7775
7	SUCCESS	7750
3	FAIL · move	2457 mm
0,2,5,6,8,9	FAIL · move	—

Selection and vision, reported as measured

INTERMEDIATE-PEAK RULE

Train past the target; score the peak, not the last checkpoint.

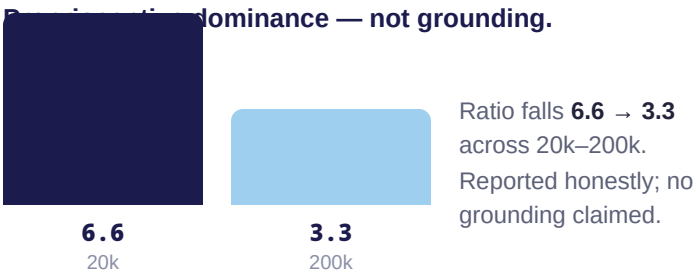
v3-100k • ~41 ép  3/10

v3-200k • ~82 ép  0/10

200k collapses to 0/10 (65–1,620 mm) — memorization degradation, filed as data.

VISION DEPENDENCE RATIO

Peak dominance — not grounding.



Place • seed 7 • 7,750 steps • scored success

40.80 ms latency · 27.85 FPS throughput

RUN (NITER 100)	LATENCY SYNC	LATENCY ASYNC	THROUGHPUT
Vehicle v3-100k FP32	40.80 ms / 24.51	40.97 / 24.39	27.85 FPS 143.66 ms
Rehearsal 20k fp32	40.83 / 24.50	40.89 / 24.43	27.22 FPS
Rehearsal 20k fp16	40.86 / 24.43	40.83 / 24.46	27.30 FPS

Host: Xeon E-2386G (12 threads) · Ubuntu 24.04 · 2026.3.0 · devices ['CPU'] · static batch-1 ([1,50,12] over 4 × 256 × 256) · forward-pass latency only.

DISCLOSURES

- **CPU-only.** No NPU on host; GPU run aborts (no iGPU).
- **FP16 ≈ FP32** (40.83 vs 40.86 ms): no-op on CPU path.
- **IR 66.658 MB**, parity 1.19e-06 (tol 1e-3).
- 20 logs + env + acceptance in-repo.

40.80 ms

MEDIAN · SYNC · N = 100

24.51 FPS

LATENCY PATH

27.85 FPS

THROUGHPUT PATH · ASYNC

Four filed ablations

ATTEMPT	MEASURED RESULT	VERDICT
1 · ACT baseline step 3000	Closed-loop 8/20 (1/4 held-out) despite 2/2 offline. Mug branch constant: var 1.20e-09 vs 1.66e-04 (×138,594). Patch → 3.0e-10, 0/8 .	KILL weights / recipe
2 · 450M VLA tune val-gated	Val 0.038 → 0.051 , stopped 14000. Best 4000 (0.025596): 1/3 offline, 0 closed-loop .	ARCHIVED negative
3 · Scripted ceiling motion probe	Sweep mean 2.6/5 , overall 0.0. Relay 7 → 8/10 vs 7/10 control. Seven grasp routes: 0.	KILL geometry
4 · Vehicle rule ~41 vs ~82 ép	3/10 vs 0/10 ; vision 6.6 → 3.3. Peak-selection + gating from filed data.	SCORED v3-100k



Grasp · mid-episode · arm A on the mug

Same scene, same commands

```
# pinned: Python 3.10 · mujoco 3.12.0 · lerobot 0.4.4 · openvino 2026.3.0
python3.10 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python scripts/env_check.py # pins
.venv/bin/python scripts/smoke_scene.py # 4 cams, headless
.venv/bin/python scripts/run_seeds.py # 10 frozen episodes
.venv/bin/python training/act_mambo_v3.py full # ACT-52M · chunk 50
.venv/bin/python scripts/eval_policy.py --checkpoint <ckpt> --seeds 0-9
.venv/bin/python scripts/export_openvino.py --checkpoint <ckpt>/pretrained_model
# Intel host only: scripts/bench_matrix_cpu.sh (niter 100)
```

LAYOUT

sim /	frozen dual-arm scene + teacher (data only)
scripts /	dataset, frozen eval, export, bench, demo
training /	ACT launchers + router contract
out /	scored IR · demo video · bench logs

Video. out/demo/MAMBO_VLA_RC_final.mp4 (10-seed cut + place clip; also seed_7.mp4).

Repo note. Contact block frozen and byte-identical across sim, training, and bench — never tuned per object, seed, or milestone.